Name: ______________________

INEQUALITIES, (REPRESENTING INEQUALITIES ON A NUMBER LINE)

Date: ________________

LO: I can represent linear inequalities on a number line using open and closed circles.

Number line representations

On a number line, an **open circle** ( ) means the value is NOT included (< or >). A **closed circle** ( ) means the value IS included (or).

Strategy: Solve the inequality first, then draw the number line with the correct circle and arrow/line.

Quick examples

●	x > 3: open circle at 3, arrow right	not including 3
●	x 5: closed circle at 5, arrow left	including 5
●	2 < x 6: open at 2, closed at 6, line between	compound
●	x > 3 uses a closed circle at 3	strict means open
●	x 5 means arrow pointing right from 5	less than means left

Key idea

Open circle = not included (strict). Closed circle = included (or equal to). Arrow shows the direction of valid values.

for < >; for

Open = excluded, Closed = included

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 – simple inequality

Represent x > 2 on a number line

Draw a number line with 2 marked

Place an open circle at 2 (not included)

Draw an arrow pointing right from 2

Open circle at 2, arrow right

The open circle shows that 2 itself is not a solution.

Example 2 – less than or equal

Represent x 4 on a number line

Draw a number line with 4 marked

Place a closed circle at 4 (included)

Draw an arrow pointing left from 4

Closed circle at 4, arrow left

The closed circle shows that 4 IS a solution.

Example 3 – compound inequality

Represent -1 < x 3 on a number line

Mark -1 and 3 on the number line

Open circle at -1 (not included), closed at 3 (included)

Draw a solid line between them

Open at -1, closed at 3, line between

No arrows needed for a compound inequality, just the connecting line.

Example 4 – solving then representing

Solve 2x + 1 > 7 and represent on a number line

2x + 1 > 7; subtract 1: 2x > 6

Divide by 2: x > 3

Open circle at 3, arrow pointing right

x > 3: open circle at 3, arrow right

Solve the inequality first, then draw the representation.

YOUR TURN – PRACTICE (deliberate practice)

1. Drawing simple inequalities

Describe the number line representation.

a) $x > 5$

b) $x \quad 2$

c) $x \quad -3$

d) $x < 0$

2. Reading number lines

Write the inequality shown.

a) Closed circle at 4, arrow right

b) Open circle at -2, arrow left

c) Open at 1, closed at 5, line between

d) Closed at -3, open at 2, line between

3. Compound inequalities

Describe the number line for each compound inequality.

a) $1 < x < 4$

b) $-2 \quad x \quad 3$

c) $0 < x \quad 5$

d) $-4 \quad x < -1$

4. Solve then represent

Solve the inequality, then describe the number line.

a) $3x > 12$

b) $x + 5 \quad 8$

c) $2x - 1 \quad 5$

d) $4x + 3 < 15$

5. Mixed problems

Solve and represent on a number line.

a) $5 \quad 2x + 1 < 11$

b) $-3 < x + 2 \quad 4$

c) $x \quad -1$ and $x < 4$

d) $3x - 2 > 1$ and $2x < 10$

e) $-4 \quad 3x + 2 \quad 11$

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick ( ) the ones that are correct. If it is wrong, write the correct answer.

a) An open circle means the value is NOT included ☐

Correct answer: __________

Reason: __________________

b) $x > 3$ uses a closed circle at 3 ☐

Correct answer: __________

Reason: __________________

c) $x \quad 5$ means closed circle at 5 with arrow pointing left ☐

Correct answer: __________

Reason: __________________

d) For $x < 4$, the arrow points right from 4 ☐

Correct answer: __________

Reason: __________________

e) A compound inequality has two endpoints and a line between them ☐

Correct answer: __________

Reason:

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) Solve **$2(x - 3) < x + 1$** and **$3x + 2 \quad 5$** simultaneously. Represent the solution on a number line and list all integer solutions.

Expression: ______________________________

Simplified: ______________________________

b) A number line shows a **closed circle at -2** and an **open circle at 4** with a line between. Write this as a double inequality. If we then apply the transformation **$y = 2x + 1$**, what inequality does y satisfy?

Expression: ______________________________

Simplified: ______________________________

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

WELL DONE!